# FEP-RLVE: Difficulty Selection as Active Inference for Reinforcement Learning on Verifiable Environments

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## Abstract

Reinforcement learning (RL) post-training of reasoning LLMs stalls when task difficulty drifts off the model's capability frontier: under group-relative objectives (GRPO/DAPO) a group of $K$ rollouts that is all-correct or all-wrong has zero advantage variance and yields *no* gradient. RL with verifiable environments (RLVE) keeps problems solvable through procedurally generated, algorithmically verified tasks with adaptive difficulty, but raises difficulty only once accuracy exceeds a threshold ($\tau_{\text{acc}}$=0.9)—a one-directional heuristic that parks the model in a low-signal regime ($U_K(0.9)$=0.34 $\ll U_K(0.5)$=0.875 at $K$=4). We recast difficulty selection as **active inference**: the trainer maintains a belief $q(s_e)$ over each environment's latent competence, updates it by minimizing *variational* free energy (a Bayesian belief update; perception), and selects the next difficulty by minimizing *expected* free energy $G_e(d) = \lambda_{\text{sig}}\,(-\ln U_K(\hat{p}(d))) - \lambda_{\text{info}}\, I(s_e; o \mid d) + \lambda_{\text{cost}}\, C(d)$, sampling $\pi(d) \propto \exp(-G_e(d)/T)$. The pragmatic term is a preference *risk* that is minimal at the signal-optimal $p$=0.5 band and diverges at degenerate difficulties; the epistemic term actively probes competence and moves difficulty in *both* directions. RLVE is the degenerate case $\lambda_{\text{info}}$=0, $T{\to}0$ with a high-accuracy preference. We implement the controller as a drop-in replacement for RLVE's scheduler and run a controlled 3-arm comparison (RLVE-90, a pragmatic-only ablation Signal-RLVE, and full FEP-RLVE) reproduced from ProRL-1.5B-v2 on a single H200. At this scale the controllers regulate difficulty as designed—FEP-RLVE holds steady-state success closest to the informative 0.5 set-point—while held-out generalization does *not* separate the arms. A mechanistic analysis attributes this to severe response truncation that makes rollout groups degenerate, so the *realized* gradient is far below the *expected* $U_K$; we report this gap as the binding constraint rather than overclaiming a generalization win.

## 1 Introduction

RL post-training of reasoning models needs a *sustained* supply of mid-difficulty experience. If problems are too hard the verifier almost always returns failure (reward sparsity); if too easy the model already solves them and the signal decays. Under group-relative policy optimization (GRPO [Shao et al., 2024], DAPO [Yu et al., 2025]) this is sharp: the advantage of a rollout is its reward centered and scaled *within its group* of $K$ samples, so a group that is unanimously correct or unanimously wrong has zero variance and contributes exactly zero gradient—DAPO discards such groups outright. The trainable signal therefore lives in *informative* groups with $0 < \sum_k o_k < K$.

RL with verifiable environments (RLVE) [Zeng et al., 2025] keeps problems in this band by generating tasks from procedural environments with algorithmic verifiers and a scalar difficulty knob, and by scaling the *number* of environments to improve generalization [SCALER Team, 2026]. Its difficulty controller, however, is a one-directional heuristic: per environment, raise the difficulty ceiling once the success rate at the ceiling exceeds $\tau_{\text{acc}}$=0.9. This parks the model at high accuracy—precisely a *low-signal* regime—never lowers difficulty if the model regresses, and has no explicit objective or exploration.

We make three contributions.

1. We show RLVE's 0.9 threshold is signal-suboptimal: the probability that a $K$-group is informative is $U_K(p) = 1 - p^K - (1-p)^K$, uniquely maximized at $p = 0.5$, so the gradient supply collapses as the controller pushes $p \to 1$ (§3, App. A).
2. We recast difficulty selection as **active inference** (Friston's Free Energy Principle [Friston, 2010, Da Costa et al., 2020]): a generative model of latent competence, a variational-free-energy belief update (perception), and an **expected free energy** objective whose pragmatic term is the preference risk $-\ln U_K$ and whose epistemic term is the information gain about competence (§3, App. B). RLVE is recovered as a degenerate special case.
3. We provide a dependency-free controller dropped into RLVE's scheduler and a controlled same-compute comparison that isolates the epistemic term (RLVE-90 / Signal-RLVE / FEP-RLVE), reported with an honest mechanistic analysis of why the arms do not separate on held-out at this scale (§4–§6).

## 2 Background

**Verifiable environments.** An RLVE environment is a procedural generator plus an algorithmic verifier exposing a scalar difficulty $d$; a sample at difficulty $d$ yields a binary correctness outcome $o \in \{0, 1\}$. **GRPO/DAPO** draw $K$ rollouts per prompt and use the group-centered reward as the advantage, so zero-variance groups carry no gradient (App. A.1). **RLVE's controller** tracks accuracy at the difficulty ceiling and bumps it when acc $\geq 0.9$; it is one-directional and threshold-based. **Active inference** [Friston, 2010, Da Costa et al., 2020] splits an agent's free-energy minimization into *perception* (update beliefs about hidden states by minimizing variational free energy = Bayesian inference) and *action* (choose policies that minimize *expected* free energy, a sum of a pragmatic preference term and an epistemic information-gain term).

## 3 Method: Difficulty Selection as Active Inference

### 3.1 Learning utility: the signal peaks at $p = 0.5$

For binary correctness with per-group success probability $p$, a group of $K$ i.i.d. rollouts is informative iff $0 < \sum_k o_k < K$, with probability

$$U_K(p) \;=\; 1 - p^K - (1-p)^K, \tag{1}$$

uniquely maximized at $p^\star = 0.5$ for all $K \geq 2$ (App. A.2), where it also maximizes the expected within-group reward variance that scales the advantages (App. A.3). Figure 1 shows the curve: at $K$=4, $U_K(0.5) = 0.875$ whereas $U_K(0.9) = 0.34$. Because success $p(d)$ decreases monotonically in difficulty $d$, there is a difficulty $d^\star$ with $p(d^\star) = 0.5$ that maximizes the expected learning signal; RLVE's rule instead targets $p \approx 0.9$.

### 3.2 Perception: a generative model of competence and its belief update

For environment $e$, let $s_e \in \mathbb{R}$ be the model's latent **competence**. Following item-response theory [Rasch, 1960], the outcome at difficulty $d$ is generated by

$$P(o = 1 \mid s_e, d) \;=\; \sigma\big(a\,(s_e - d)\big), \tag{2}$$

with discrimination $a$. The trainer maintains a belief $q(s_e)$ (a grid over $s_e$); predicted success is $\hat{p}(d) = \mathbb{E}_{s\sim q}[\sigma(a(s-d))]$. Given a batch of outcomes, the *variational free energy*

$$F[q] = \mathbb{E}_q[\ln q(s) - \ln p(o, s)] = D_{\mathrm{KL}}\big(q(s) \,\|\, p(s \mid o)\big) - \ln p(o) \tag{3}$$

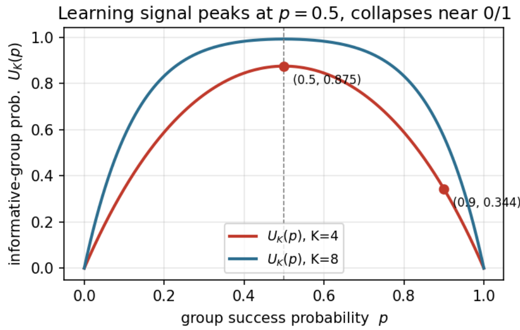


Figure 1: Informative-group probability $U_K(p) = 1 - p^K - (1-p)^K$. The learning signal is maximal at $p = 0.5$ and collapses as $p \to 0$ or 1. RLVE's acc $\geq 0.9$ operating point ($U_K$=0.34 at $K$=4) is far down the right shoulder.

is minimized by the Bayesian posterior, i.e. the exact grid update $q(s) \leftarrow q(s) \prod_k P(o_k \mid s, d_k)/Z$ (perception; App. B.2). The belief is maintained *per environment* and updated only from that environment's own outcomes.

### 3.3 Action: expected free energy over difficulties

To choose the next difficulty, active inference minimizes the expected free energy

$$G_e(d) \;=\; \underbrace{\lambda_{\text{sig}}\big(-\ln U_K(\hat{p}(d))\big)}_{\text{pragmatic / preference risk}} \;-\; \underbrace{\lambda_{\text{info}}\, I_q(s_e; o \mid d)}_{\text{epistemic / info gain}} + \lambda_{\text{cost}}\, C(e, d). \tag{4}$$

The preference is over *outcomes*, and crucially is not "always correct" (which would prefer easy tasks) but "the group is *informative*", whose predicted probability is $U_K(\hat{p}(d))$; the preference risk $\mathbb{E}_{q(o|d)}[-\ln P_{\text{pref}}(o)] = -\ln U_K(\hat{p}(d))$ is minimal at $p$=0.5 and diverges as $U_K \to 0$, strongly repelling degenerate (all-pass/all-fail) difficulties (App. B.3). The epistemic term is the mutual information $I_q(s_e; o \mid d) = H[\text{Ber}(\hat{p})] - \mathbb{E}_{s\sim q} H[\text{Ber}(\sigma(a(s-d)))]$, largest where outcomes are most diagnostic of competence; its location tracks the current belief uncertainty rather than a fixed 0.5 target, letting the controller actively probe the frontier. The policy is the active-inference softmax

$$\pi(d \mid e) \;\propto\; \exp\big(-G_e(d)/T\big), \tag{5}$$

greedy as $T \to 0$ and uniform as $T \to \infty$. **RLVE is the degenerate case** $\lambda_{\text{info}}$=0, $T \to 0$, with a high-accuracy (rather than informative-band) preference (App. B.5). Setting only $\lambda_{\text{info}}$=0 gives our *Signal-RLVE* ablation, which targets $p$=0.5 but does not probe competence.

### 3.4 Online controller

The belief is a per-environment grid (exact Bayes; no Gaussian/Laplace approximation); `observe()` performs the update, `G()/policy()` score and sample difficulties. The controller is dependency-free and self-checked ($U_K$ and information gain both peak at $p$=0.5; belief $\sigma$ contracts as competence rises). It is wired into RLVE's SLIME rollout scheduler, replacing the acc $\geq 0.9$ bump and leaving the RL core (GRPO) unchanged.

## 4 Experimental setup

We reproduce RLVE on the SLIME backend from **ProRL-1.5B-v2** (`Nemotron-Research-Reasoning-Qwen-1.5B`) [Liu et al., 2025] on a single H200 (the official 8-GPU recipe reconfigured to one card; see `rlve_repro/`). Training uses 4 environments for 30 rollout steps with $K$=4 samples per

Table 1: Three difficulty controllers, same compute (ProRL-1.5B-v2, 4 envs, 30 steps, $K$=4). $p$ is mean training success; $U_K$ the mean effective sample ratio; "last-5 $p$" the steady-state success; held-out is the final reward on 128 instances with its change $\Delta$ from initialization. Bold marks the best held-out, but see §6: the held-out differences are within run-to-run noise.

| Controller | mean $p$ | mean $U_K$ | last-5 $p$ | held-out | $\Delta$ |
|---|---|---|---|---|---|
| RLVE-90 (acc$\geq$0.9) | 0.552 | 0.853 | 0.519 | $-0.891$ | +0.016 |
| Signal-RLVE ($\lambda_{\text{info}}$=0) | 0.491 | 0.852 | 0.444 | $\mathbf{-0.813}$ | **+0.109** |
| FEP-RLVE (ours) | 0.477 | 0.850 | **0.531** | $-0.875$ | +0.031 |

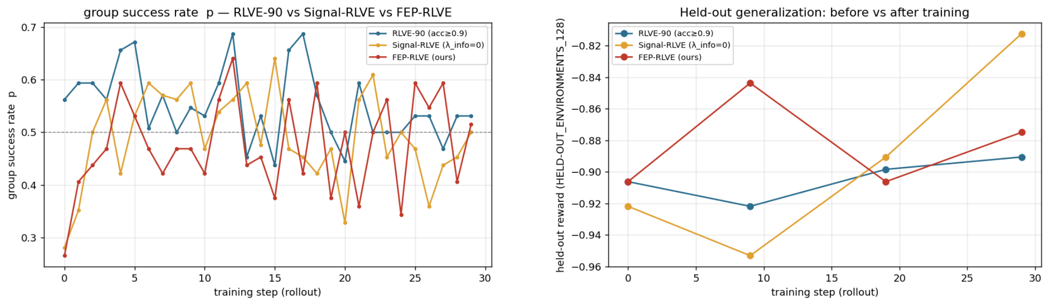


Figure 2: **Left:** training group success vs. step; the EFE arms hold near the 0.5 set-point (dashed) while RLVE-90 sits higher. **Right:** held-out reward before vs. after training—flat and near-floor for all arms; differences are within noise.

prompt (rollout batch 8, oversample 8), difficulty range $[0, 16]$, controller temperature $T$=0.25, $\lambda_{\text{sig}}$=1, $\lambda_{\text{cost}}$=0, response budget 2048 tokens. We evaluate on a fixed held-out set of 128 environment instances at steps $0, 9, 19, 29$. Three arms share the *same* recipe and differ *only* in the difficulty controller: **RLVE-90** (the acc $\geq$ 0.9 baseline), **Signal-RLVE** ($\lambda_{\text{info}}$=0), and **FEP-RLVE** ($\lambda_{\text{info}}$=1). Signal-RLVE is the key ablation: only if FEP-RLVE beats it is the *epistemic* term doing work, rather than "just move 0.9 → 0.5." Metrics are the training group success $p$, the effective sample ratio $U_K(p)$, and held-out reward before vs. after.

## 5 Results

**Difficulty regulation (works as designed).** Both EFE arms hold training success near the informative band, and FEP-RLVE parks *steady-state* success (last-5 mean 0.531) closest to the $p$=0.5 set-point, consistent with the Gibbs-optimal theory (Table 1, Fig. 2 left). The effective sample ratio is essentially identical across arms ($\approx 0.85$): at $K$=4 with $p$ near 0.5, $U_K$ is flat, so this metric cannot by itself separate the controllers—a point we return to below.

**Generalization does not separate the arms.** Held-out reward is flat and near the floor for all three arms throughout training ($\approx -0.81$ to $-0.95$, i.e. 3–9% success; Fig. 2 right). Signal-RLVE shows the largest nominal improvement ($\Delta = +0.109$), but on 128 instances this is $\approx 5$ additional solved problems—about 1.7 binomial standard errors from a single seed with four noisy eval points—and we therefore do not read it as a real ranking (§6).

**Mechanism: why the arms do not separate.** Process diagnostics (Fig. 3) tell a consistent story. (i) Response *truncation* is pinned at 0.7–0.9 throughout training (and

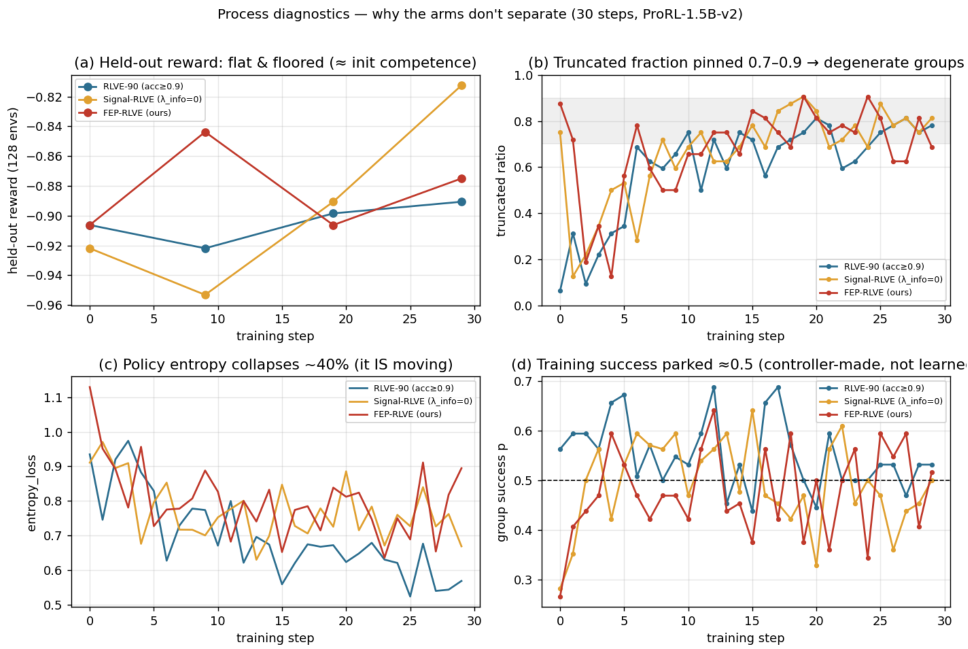


Figure 3: Process diagnostics. (a) held-out is flat/floored for all arms; (b) truncation pinned 0.7–0.9; (c) policy entropy collapses $\approx 40\%$; (d) training success parked near 0.5 (controller-made, not learned).

$\approx 0.98$ at eval), so the model frequently hits the 2048-token budget before emitting a verifiable answer. (ii) With truncation = failure, many $K{=}4$ groups become *degenerate* (all samples truncate/fail), so the realized advantage variance collapses and the policy-gradient loss stays at $\sim 2 \times 10^{-4}$—orders of magnitude below a healthy run. (iii) Policy entropy nonetheless drops $\approx 40\%$, so the model commits without acquiring transferable skill, and held-out stays floored. Crucially, training success $\approx 0.5$ is therefore *manufactured by the controller selecting solvable difficulties*, not by the model improving: a flat 0.5 curve is evidence the controller works, not that capability rose.

## 6 Discussion and limitations

**Expected vs. realized signal.** $U_K(p)$ assumes the $K$ samples in a group are *independent* Bernoulli($p$) draws of correctness, giving $U_K \approx 0.85$ here. Truncation breaks this: when most completions are cut off, group outcomes become correlated and degenerate, so the *realized* informative fraction is far below the *expected* $U_K$. The effective-ratio metric, computed from the same closed form, is blind to this degeneracy by construction. This gap—not the choice of controller—is the binding constraint at this scale, and it is the clearest target for future work (raise the response budget; decouple "truncated" from "wrong" in the reward; feed the *measured* non-degenerate-group fraction back into the utility so the controller optimizes realized rather than expected informativeness). **What this study does and does not establish.** It validates that the EFE controller regulates difficulty as designed, and that higher training success (RLVE-90) does *not* imply better generalization—the motivating claim that accuracy is the wrong curriculum signal. It does *not* establish a generalization win for any arm: 30 steps at 1.5B with a floored, truncation- suppressed held-out set leave no room for transfer to appear, and the held-out gaps are within single-seed noise. **Other limitations.** Competence is modeled as a single scalar $s_e$ per environment (real skills are multi-dimensional and shared across environments); environments are modeled independently, ignoring transfer. When the model outgrows $d_{\max}$ all levels saturate ($U_K \to 0$ everywhere) and *no* controller has signal, motivating environment scaling [Zeng et al., 2025, SCALER Team, 2026].

## 7 Related work

RL from verifiable rewards [Lambert et al., 2024] and its environment-scaling form RLVE [Zeng et al., 2025] and SCALER [SCALER Team, 2026] regulate difficulty toward the capability frontier; we replace their accuracy heuristic with an explicit free-energy objective. Group-relative RL (GRPO [Shao et al., 2024], DAPO [Yu et al., 2025]) motivates the informative-group utility; ProRL [Liu et al., 2025] is our base model. Our formulation is an instance of active inference / the Free Energy Principle [Friston, 2010, Da Costa et al., 2020], and the epistemic term connects to learning-progress automated curricula [Graves et al., 2017], which it generalizes through an explicit belief over competence.

## 8 Conclusion

Casting difficulty selection as active inference replaces RLVE's accuracy-threshold heuristic with a principled objective—a Bayesian belief over competence plus an expected-free-energy criterion—that targets the signal-optimal operating point, actively probes competence, and moves difficulty both ways, at near-zero extra compute and with the RL core unchanged. In a small-scale reproduction the controller regulates difficulty as designed; honest diagnostics show that response truncation, not the controller, currently caps learning by collapsing the realized gradient below the expected one. Closing that expected-vs-realized gap is the natural next step.

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## A Why the learning signal is maximized at $p = 0.5$

**A.1 Vanishing advantages on zero-variance groups.** GRPO/DAPO set $A_j = (r_j - \bar{r})/(\sigma_r + \varepsilon)$ with $\bar{r} = k/G$ and $\sigma_r^2 = \hat{p}(1-\hat{p})$ for $k$ correct of $G$. If $k \in \{0, G\}$ then $\sigma_r = 0$ and all $A_j = 0$: no gradient. **A.2 Informative-group probability.** $P_{\text{inf}}(p) = 1-(1-p)^G - p^G$; $P'_{\text{inf}}(p) = G[(1-p)^{G-1} - p^{G-1}] = 0 \Rightarrow p^\star = 1/2$, a maximum for all $G \geq 2$. **A.3 Variance.** $\text{Var}[r] = p(1-p)$ peaks at 0.5 and the expected empirical variance $p(1-p)(1-1/G)$ shares the maximizer, so both the *frequency* of informative groups and the *magnitude* of their advantages peak at $p = 0.5$. **A.4 Consequence.** Since $p(d)$ is decreasing in $d$, some $d^\star$ gives $p = 0.5$; RLVE's $\tau_{\text{acc}}$=0.9 keeps the model at low $U_K$ and never lowers difficulty. For shaped rewards the optimum shifts off 0.5 to wherever $\text{Var}[r]$ is maximal and $p^\star$ becomes a tunable target.

## B Active-inference formulation

This is Friston's Free Energy Principle, not a thermodynamic energy−entropy objective. **B.1 Generative model:** $P(o{=}1 \mid s_e, d) = \sigma(a(s_e - d))$, belief $q(s_e)$, $\hat{p}(d) = \mathbb{E}_{s\sim q}[\sigma(a(s-d))]$. **B.2 Perception:** minimizing $F[q] = D_{\text{KL}}(q(s)\|p(s \mid o)) - \ln p(o)$ gives the exact grid posterior $q(s) \leftarrow q(s)\prod_k P(o_k \mid s, d_k)/Z$. **B.3 Pragmatic term:** preferring informative outcomes gives risk $-\ln U_K(\hat{p})$, which diverges as $U_K \to 0$. **B.4 Epistemic term:** $I_q(s_e; o \mid d) = H[\text{Ber}(\hat{p})] - \mathbb{E}_{s\sim q} H[\text{Ber}(\sigma(a(s-d)))]$. **B.5 Policy:** $\pi(d \mid e) \propto \exp(-G_e(d)/T)$; RLVE is $\lambda_{\text{info}}$=0, $T \to 0$, high-accuracy preference.

## C Reproduction

Single-GPU RLVE reproduction steps and the controller integration point are in `rlve_repro/` (`active_inference_controller.py`, `active_inference_manager.py`); analysis and figures are regenerated by `analyze_rlve.py` and `diagnostics.py`.